

Asia PCB, CCL, Substrate, Testing, and Passive Components

Jerry Tsai ^{AC}

jerry.tsai@jpmorgan.com

J.P. Morgan Securities (Taiwan) Limited

(886-2) 2725-9867

CCL/PCB

AI driving HPC CCL/PCB TAM growth

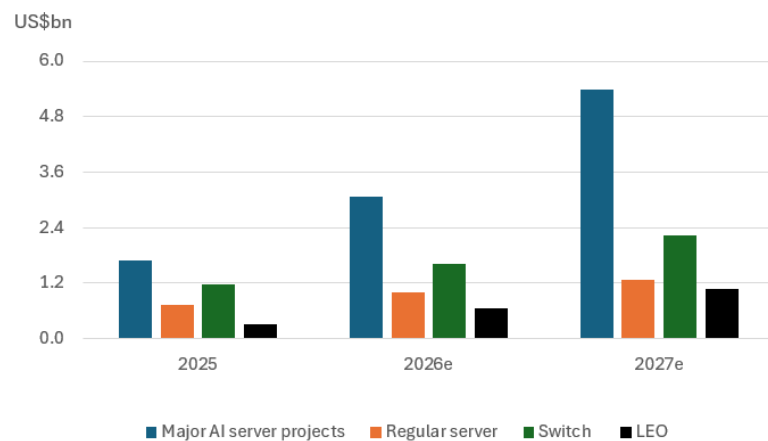
HPC CCL TAM

(US\$bn)	2025	2026e	2027e	YoY growth %	2026e	2027e	2025-27 CAGR	Mix %	2025	2026e	2027e
Major AI server projects	1.69	3.07	5.39	Major AI server projects	82%	75%	79%	Major AI server projects	43%	49%	54%
Regular server	0.73	0.99	1.28	Regular server	36%	29%	32%	Regular server	19%	16%	13%
Switch	1.17	1.62	2.24	Switch	39%	39%	39%	Switch	30%	26%	22%
LEO	0.30	0.65	1.07	LEO	116%	65%	89%	LEO	8%	10%	11%
Total	3.89	6.33	9.98	Total	63%	58%	60%	Total	100%	100%	100%

(US\$bn)	2025	2026e	2027e	YoY growth %	2026e	2027e	2025-27 CAGR	Mix %	2025	2026e	2027e
M9	-	0.24	0.77	M9		214%		M9	0%	4%	8%
M8	1.59	3.66	6.49	M8	130%	77%	102%	M8	41%	58%	65%
M7 and below	2.30	2.43	2.72	M7 and below	6%	12%	9%	M7 and below	59%	38%	27%
Total	3.89	6.33	9.98	Total	63%	58%	60%	Total	100%	100%	100%

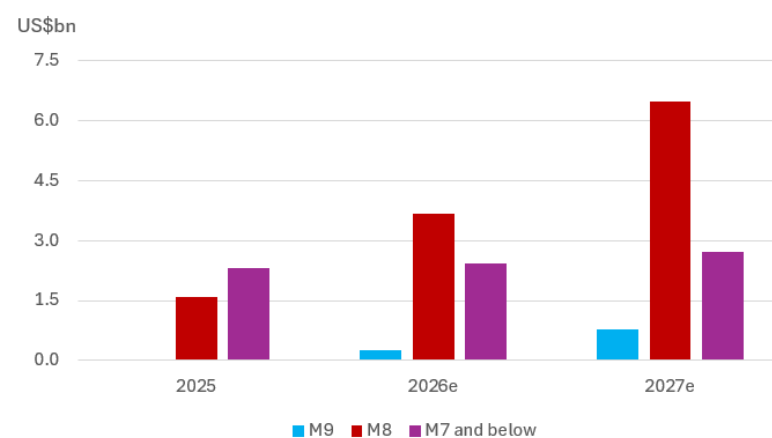
Source: J.P. Morgan estimates

HPC CCL TAM by application



Source: J.P. Morgan estimates.

TAM by CCL grade



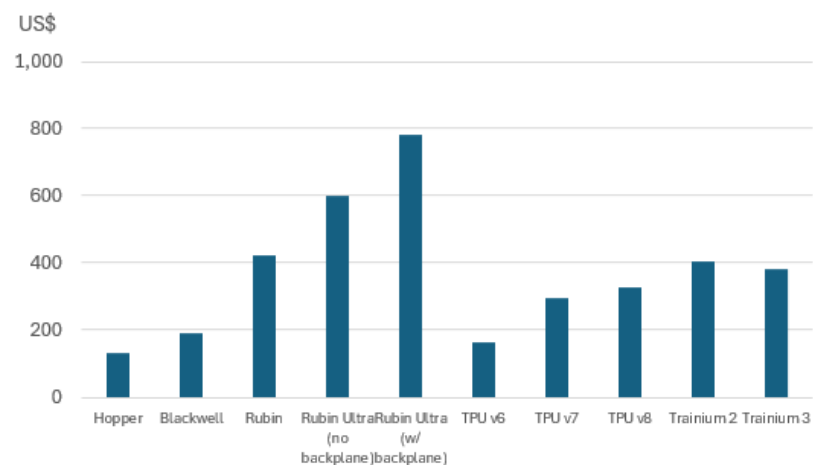
Source: J.P. Morgan estimates.

J.P.Morgan

Not

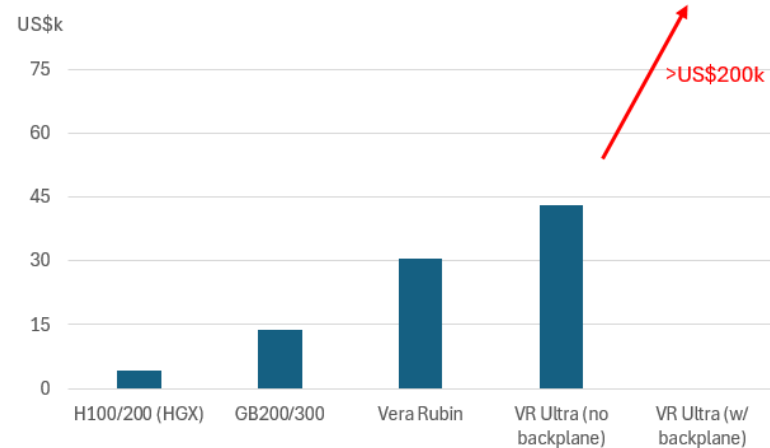
HPC CCL/PCB: massive content growth in the foreseeable future

CCL content per AI chip



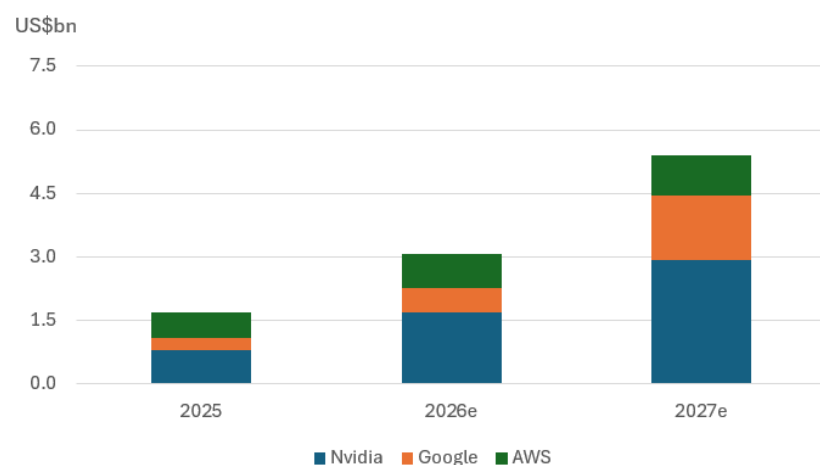
Source: J.P. Morgan estimates.

CCL content per AI rack



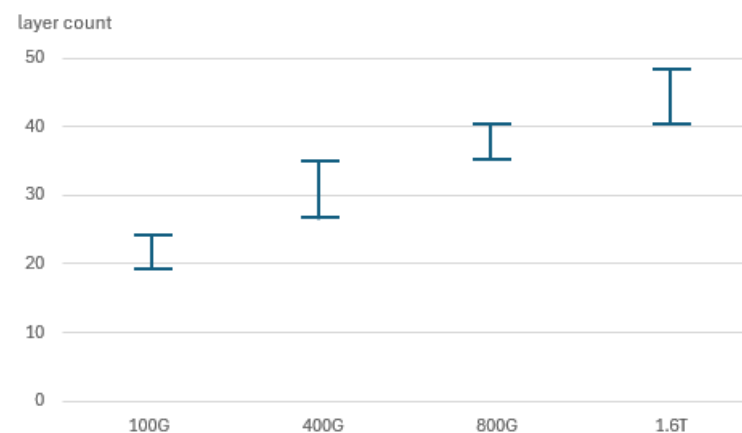
Source: J.P. Morgan estimates.

CCL TAM for Nvidia/Google/AWS



Source: J.P. Morgan estimates.

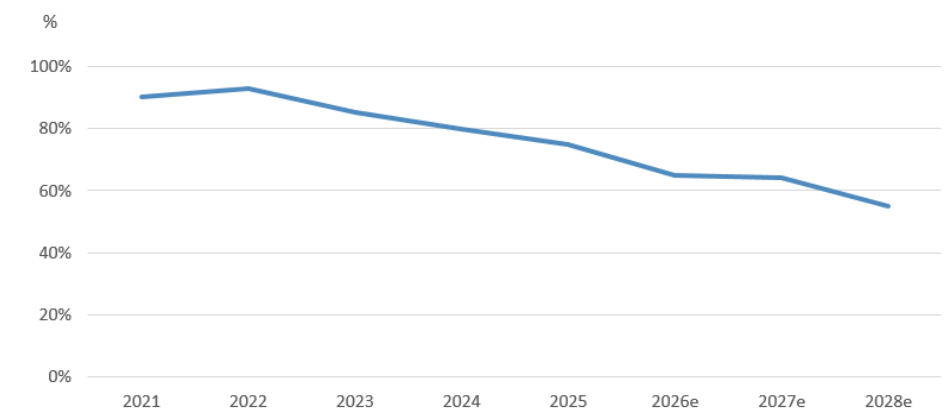
PCB layer count comparison (switch as illustration)



Source: J.P. Morgan estimates.

HPC CCL/PCB: supply could undershoot demand

PCB/HDI fabrication yield



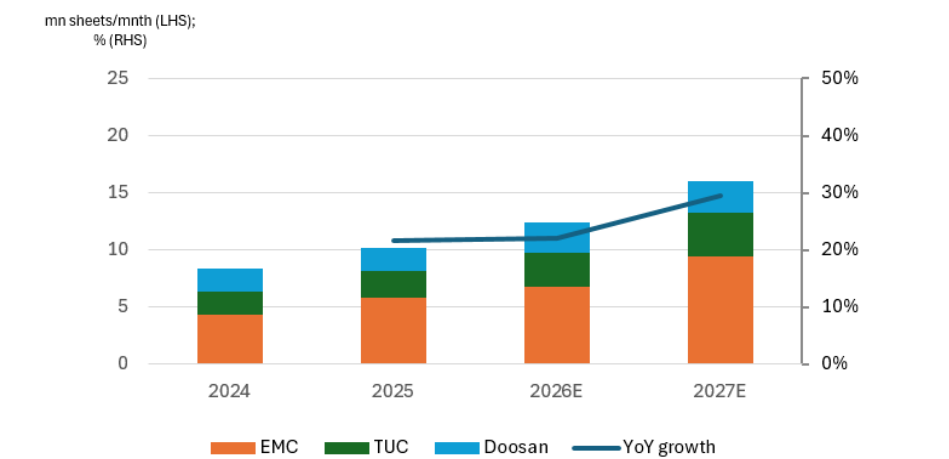
Source: J.P. Morgan estimates.

M9 projects

M9 projects		
Vera Rubin generation	Midplane	
	CPX	Potential to go to M10
Vera Rubin Ultra generation	Midplane	
	CPX	Potential to go to M10
	LPU/LPX	
	Blackplane	Potential to go to M10

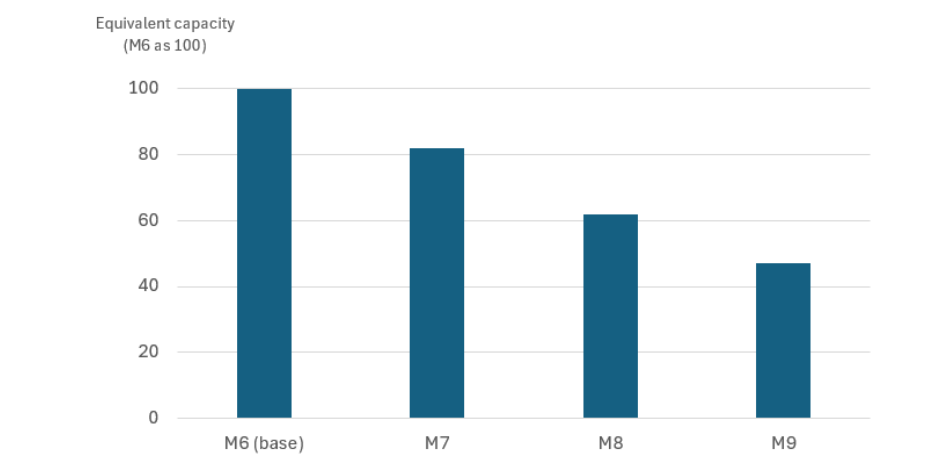
Source: J.P. Morgan estimates.

CCL capacity growth



Source: J.P. Morgan estimates, company data.

CCL capacity conversion loss



Source: J.P. Morgan estimates.

HPC CCL/PCB: regular server and LEO under strong growth

Regular server CCL/PCB TAM

Server CPU Shipment (K)	2020	2021	2022	2023	2024	2025	2026e	2027e
AMD	2,022	3,837	5,761	5,256	5,820	6,219	7,125	7,512
Intel	22,026	20,066	19,181	14,375	14,783	13,492	12,959	12,509
PCB ASP (US\$)	2020	2021	2022	2023	2024	2025	2026e	2027e
Intel	40	59	96	148	187	227	285	322
AMD	42	69	112	157	218	253	360	400
Blended ASP	37	50	75	111	140	166	202	224
Blended ASP YoY %		36%	49%	49%	26%	18%	22%	11%
Server PCB TAM (US\$bn)	2020	2021	2022	2023	2024	2025	2026e	2027e
	0.46	0.65	1.03	1.26	1.69	1.98	2.68	3.46
TAM YoY %		39%	60%	22%	34%	17%	36%	29%
Server CCL TAM (US\$bn)	2020	2021	2022	2023	2024	2025	2026e	2027e
Before yield	0.15	0.22	0.34	0.42	0.56	0.66	0.89	1.15
Yield	90%	90%	90%	90%	90%	90%	90%	90%
After yield	0.17	0.24	0.38	0.47	0.63	0.73	0.99	1.28

Source: J.P. Morgan estimates.

J.P.Morgan

HPC CCL/PCB: regular server and LEO under strong growth

LEO CCL/PCB TAM

Terminal					
	2023	2024	2025	2026e	2027e
User terminal PCB TAM (NT\$mn)	4.09	9.41	20.25	45.80	82.94
User terminal PCB TAM (US\$mn)	0.14	0.31	0.68	1.53	2.76
YoY		130%	115%	126%	81%
Satellite					
	2023	2024	2025	2026e	2027e
Satellite PCB TAM (NT\$mn)	3.88	4.13	6.79	12.55	13.24
Satellite PCB TAM (US\$mn)	0.13	0.14	0.23	0.42	0.44
UT + satellite					
	2023	2024	2025	2026e	2027e
PCB TAM (US\$bn)	0.27	0.45	0.90	1.95	3.21
YoY		70%	100%	116%	65%
	2023	2024	2025	2026e	2027e
CCL TAM (US\$bn)	0.09	0.15	0.30	0.65	1.07
YoY		70%	100%	116%	65%

Source: J.P. Morgan estimates, company data.

PCB/CCL: peer valuation

Valuations

Company			Price (LCY\$)	PE (x)				PB (x)				NI yoy (%)			ROE (%)		
Ticker		2024		2025	2026E	2027E	2024	2025	2026E	2027E	2025	2026E	2027E	2025	2026E	2027E	
CCL																	
	Elite Material	2383 TT	3,220	115.8	77.3	43.4	26.8	31.8	22.9	17.4	12.3	53%	81%	62%	34%	46%	54%
	Taiwan Union	6274 TT	797	83.4	65.7	34.9	23.0	15.3	12.4	10.8	9.1	31%	93%	58%	21%	33%	40%
	Iteq	6213 TT	217	95.8	52.0	32.7	23.4	3.8	3.7	3.6	3.4	84%	60%	39%	7%	11%	14%
	Doosan Co Ltd	000150 KS	1,264,000	N.A.	298.8	44.0	29.4	11.7	11.2	12.3	9.0	N.A.	524%	52%	4%	23%	27%
	Shengyi Tech C-A	600183 CH	59	80.0	40.9	26.4	19.5	9.6	8.3	7.0	5.8	103%	56%	37%	21%	27%	31%
PCB																	
	Gold Circuit	2368 TT	1,095	94.9	56.2	33.3	22.6	25.0	16.4	13.4	10.2	71%	79%	48%	35%	49%	55%
	Victory Giant -A	300476 CH	278	207.5	55.5	26.8	16.3	26.9	15.5	10.3	7.0	274%	110%	64%	34%	41%	43%
	Isupetasys Co	007660 KS	117,000	102.7	51.4	30.4	20.1	23.2	11.4	8.3	5.9	117%	76%	51%	30%	31%	34%
	Wus Printed-A	002463 CH	86	63.5	43.2	30.7	21.8	13.9	10.9	8.8	6.7	48%	47%	40%	28%	30%	32%
	Ttm Technologies	TTMI US	108	195.5	62.2	33.9	24.5	7.0	6.3	6.2	5.5	215%	94%	39%	11%	16%	19%
	Shennan Circui-A	002916 CH	243	86.3	49.5	32.6	23.4	11.1	9.4	7.3	5.8	74%	55%	37%	21%	26%	28%
	Shengyi Electr-A	688183 CH	89	222.5	45.5	27.7	20.2	17.3	12.9	9.3	6.8	390%	64%	37%	30%	36%	37%
	Shenzhen Kinwo-A	603228 CH	59	44.1	45.4	26.5	19.0	4.8	4.5	3.9	3.4	5%	80%	36%	10%	16%	19%
	Dynamic Holding	3715 TT	187	49.6	69.3	32.8	10.9	6.6	4.9	4.7	4.2	-29%	118%	198%	8%	19%	N.A.
	Tripod Tech Corp	3044 TT	378	23.7	19.4	14.0	10.6	4.0	3.7	3.2	2.8	22%	38%	33%	20%	24%	28%
	Delton Technol-A	001389 CH	127	76.6	53.0	35.3	23.2	17.6	13.6	10.8	7.1	50%	63%	53%	29%	31%	33%
	Taiwan Pcb	8213 TT	34	12.2	14.8	N.A.	N.A.	0.6	0.6	N.A.	N.A.	-18%	N.A.	N.A.	4%	N.A.	N.A.
	Hannstar Board	5469 TT	85	14.4	14.1	N.A.	N.A.	1.3	1.1	N.A.	N.A.	-3%	N.A.	N.A.	9%	N.A.	N.A.
	Compeq Mfg	2313 TT	282	59.9	51.1	36.2	24.5	7.5	7.0	6.1	5.1	17%	41%	48%	14%	18%	23%
	Unitech Print	2367 TT	78	32.8	114.0	39.3	39.7	4.3	4.3	3.8	N.A.	-69%	306%	-30%	4%	N.A.	N.A.
	Meiko Electronic	6787 JT	27,020	47.4	34.2	29.2	22.8	6.4	5.5	4.7	4.0	36%	17%	28%	18%	19%	22%
	Zhen Ding Tech	4958 TT	264	27.2	38.1	21.4	14.8	2.3	2.3	2.0	N.A.	-26%	85%	49%	6%	9%	13%
	Suzhou Dongsha-A	002384 CH	132	206.1	122.2	42.6	25.3	12.0	11.3	9.0	6.7	73%	211%	47%	9%	23%	28%
	Cmk Corp	6958 JT	592	11.1	13.2	22.2	17.6	0.5	0.5	0.5	0.5	-15%	-41%	26%	3%	2%	3%
	Flexium Intercon	6269 TT	66	N.A.	N.A.	N.A.	N.A.	0.9	1.0	N.A.	N.A.	N.A.	N.A.	N.A.	-10%	N.A.	N.A.
	Chin-Poon Ind	2355 TT	53	18.6	33.1	29.8	19.3	1.2	1.2	1.2	1.1	-44%	21%	41%	4%	5%	7%

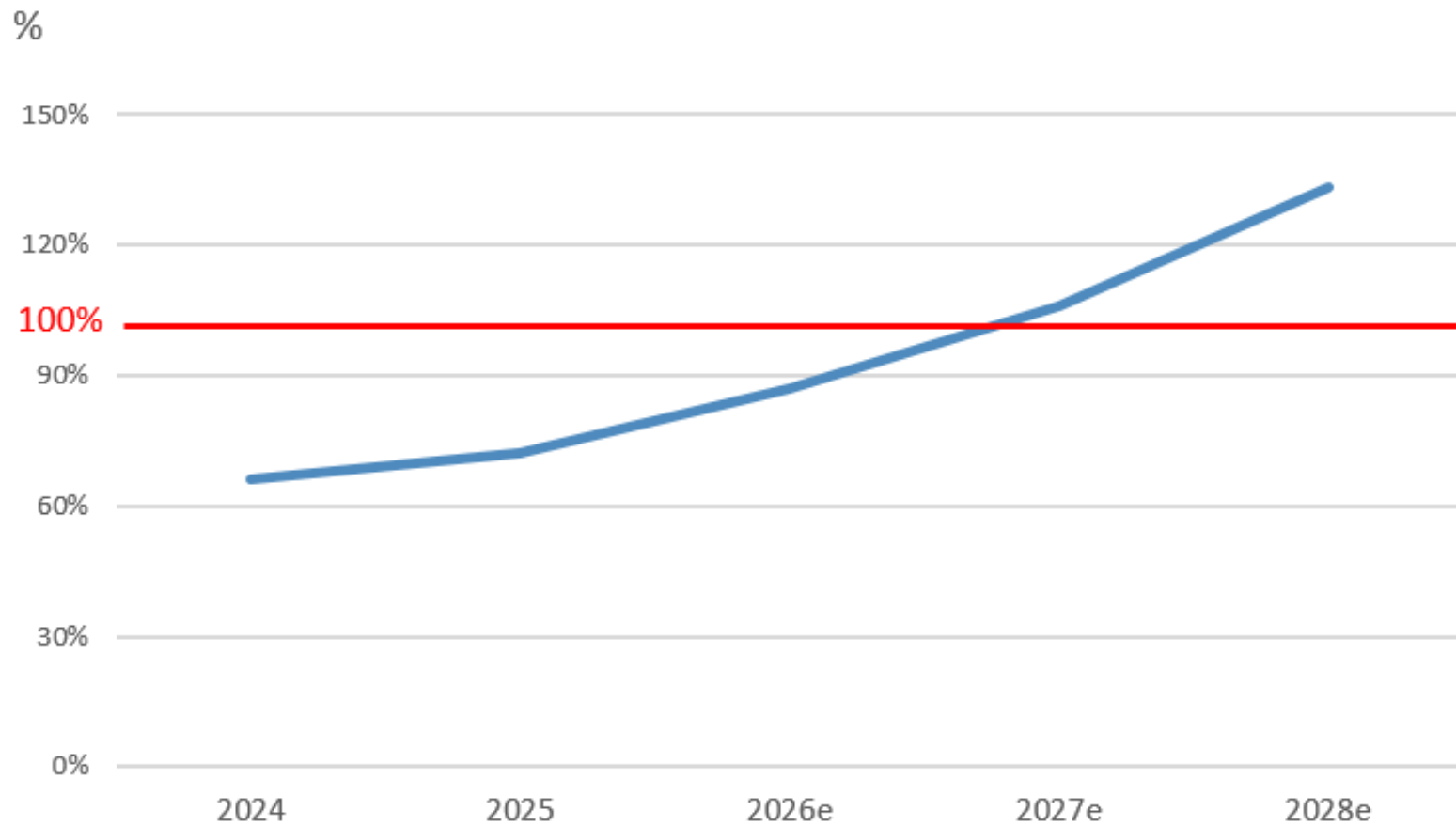
Note: 1) Share prices as of April 9; 2) J.P. Morgan covers Elite Material (covered by Jerry Tsai), Isupetasys (by Sangsik Lee), Tripod Tech (by Jerry Tsai), and Flexium (by Jerry Tsai), whose financial forecasts are based on J.P. Morgan estimates. Others are based on Bloomberg Finance L.P. estimates.

Source: J.P. Morgan estimates, company data, Bloomberg Finance L.P.

Substrate

Substrate: entering unprecedented supply tightness

Industry UTR trend



Source: J.P. Morgan estimates.

J.P.Morgan

Substrate: entering unprecedented supply tightness

Global substrate demand breakdown

Demand Summary

(bn mm ²)	2024	2025	2026e	2027e	2028e
AI server & switch	2,448	4,634	8,301	13,638	21,522
Regular server	3,210	3,243	3,682	3,911	4,183
Consumer	3,812	3,431	2,941	3,006	3,055
Total	9,470	11,308	14,924	20,555	28,759

YoY %

	2025	2026e	2027e	2028e	2025-28 CAGR
AI server & switch	89%	79%	64%	58%	67%
Regular server	1%	14%	6%	7%	9%
Consumer	-10%	-14%	2%	2%	-4%
Total	19%	32%	38%	40%	36%

Demand Mix

%	2024	2025	2026e	2027e	2028e
AI server & switch	26%	41%	56%	66%	75%
Regular server	34%	29%	25%	19%	15%
Consumer	40%	30%	20%	15%	11%
Total	100%	100%	100%	100%	100%

Source: Company data, J.P. Morgan estimates.

Substrate: massive size migration ahead

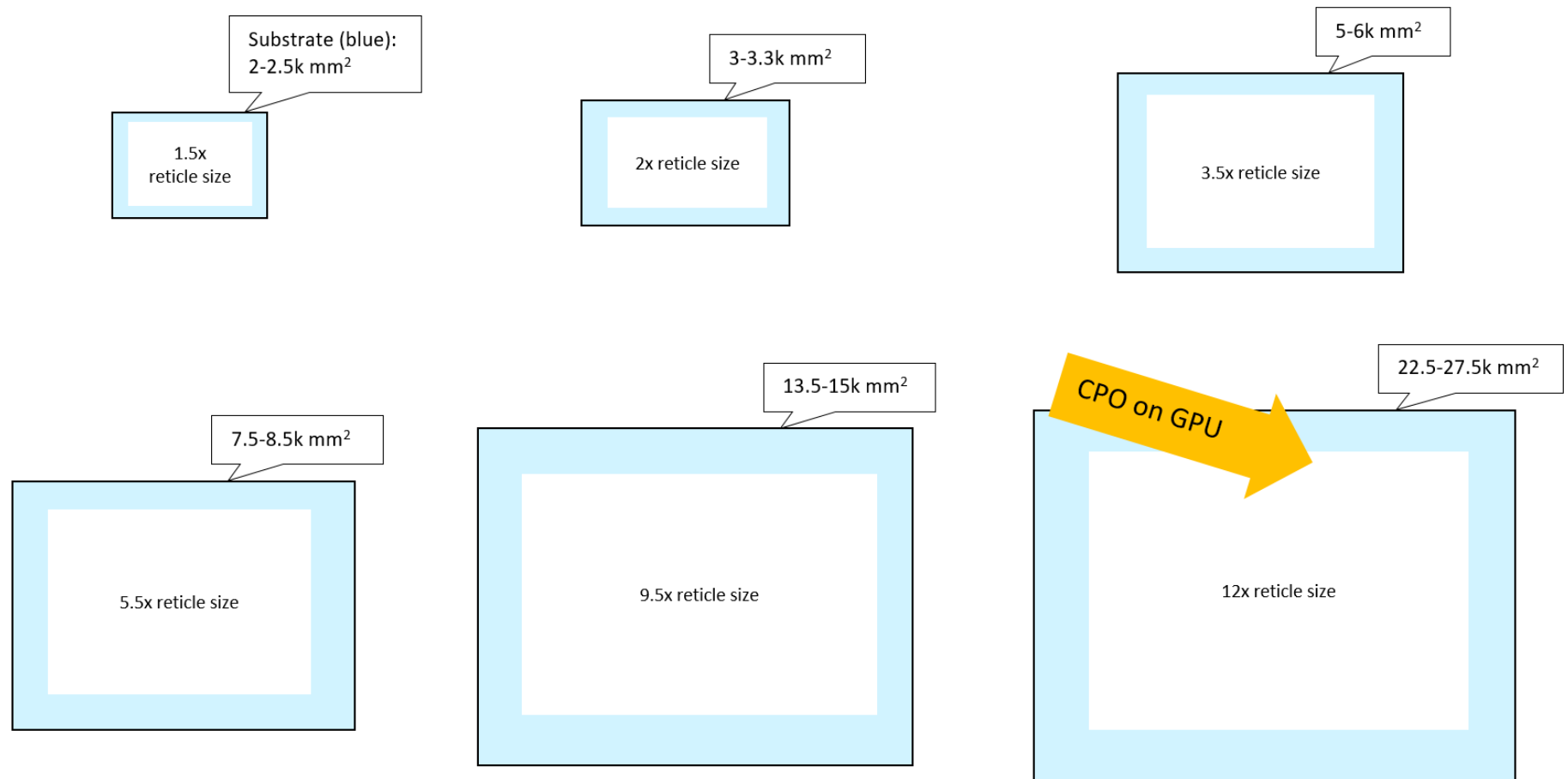
Major AI chip body size & layer count

Vendor	AI chip	Body size (mm ²)	Layer count	Total size (mm ²)
Nvidia	Hopper	3,190	12	38,280
	Grace	4,964	12	59,568
	Blackwell	5,913	14	82,782
	Rubin	8,051	18	144,918
	Rubin Ultra	16,102	20	322,040
	Feynman	18,000	24	432,000
	NVLink 5	3,969	16	63,504
	NVLink 6	4,500	18	81,000
	NVLink 7	5,500	20	110,000
	NVLink 8 (CPO)	8,000	22	176,000
Google (TPU)	Trillium (v6)	3,100	12	37,200
	Hellcat (v7) (BRCM)	3,460	12	41,520
	Sunfish (v8) (BRCM)	7,037	18	126,666
	Pumafish (v9) (BRCM)	18,000	24	432,000
AWS	Trainium 2	5,514	14	77,200
	Trainium 3	6,344	18	114,188
	Trainium 4	14,400	24	345,600

Source: J.P. Morgan estimates.

Substrate: massive size migration ahead

CoWoS interposer size vs substrate



Source: J.P. Morgan estimates.

Substrate: entering unprecedented supply tightness

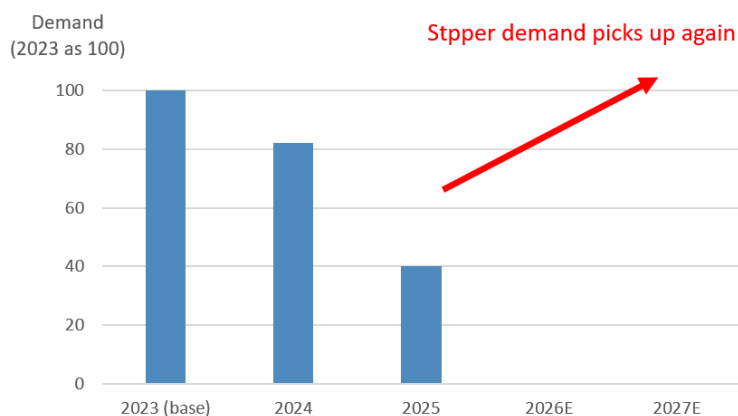
Global substrate supply

(equivalent to 30x30mm, 6 layers)

Capacity	2018	2019	2020	2021	2022	2023	2024	2025	2026e	2027e	2028e
Unimicron	43	51	55	58	67	69	73	79	92	102	117
Ibiden	32	34	44	55	60	63	65	72	82	95	107
Nanya	28	29	31	36	42	45	47	48	50	54	57
SEMCO	17	18	19	20	22	24	30	32	33	36	40
Shinko	24	24	30	35	41	44	46	49	50	53	55
Kinsus	6	8	12	16	21	26	26	27	28	31	37
AT&S	10	11	14	21	32	35	38	40	42	45	49
Others	10	11	13	13	15	20	23	26	30	34	41
Total	170	186	218	254	300	326	348	373	407	449	503
Growth		9.1%	17.3%	16.5%	18.3%	8.7%	6.8%	7.0%	9.3%	10.2%	12.1%
<i>Absolute growth</i>		16	32	36	46	26	22	24	35	42	54

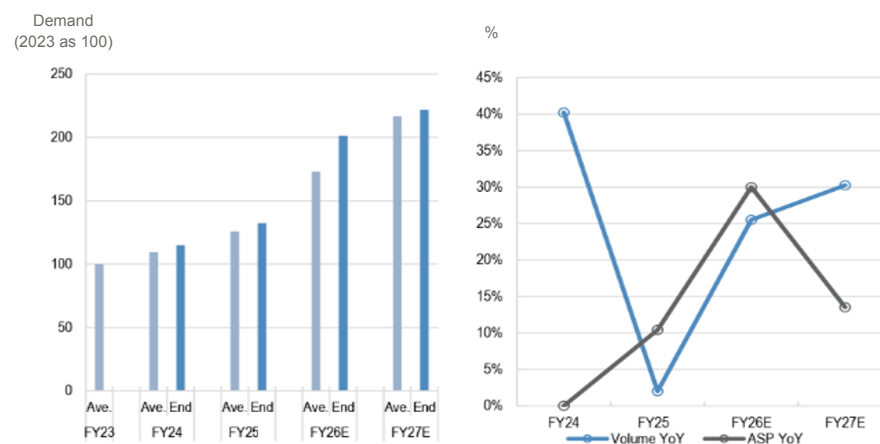
Source: Company data, J.P. Morgan estimates.

Global stepper demand forecast



Source: Company data, J.P. Morgan estimates.

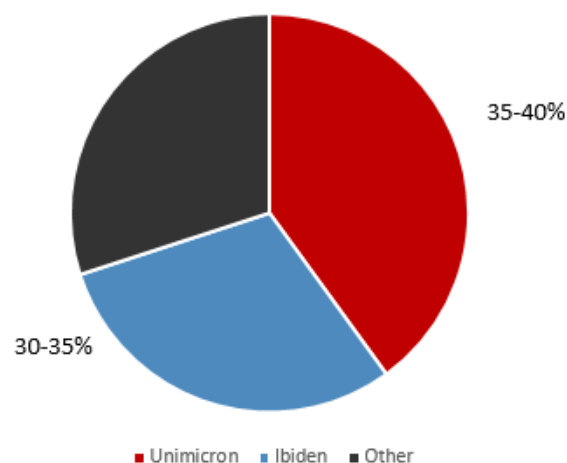
Nittobo T-glass capacity expansion (left) and vol/pricing trend (right)



Source: J.P. Morgan estimates, company data.

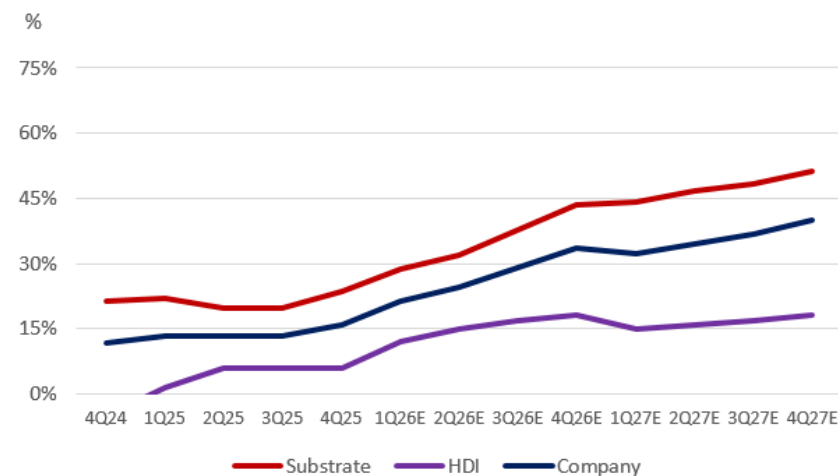
Substrate: positive implication from T-glass shortage; LTA emerging

T-glass allocation



Source: J.P. Morgan estimates

Unimicron margin forecasts



Source: J.P. Morgan estimates, company data.

LTA comparison by type

LTA types	Financial benefits	Flexibility	Irrevocability
Pre-paid sales	*	**	****
Customer-purchased equipment	**	**	****
Customer-financed capacity	***	*	*****
Capacity reservation fees	*****	****	****
Super hot runs	****	****	****

Source: J.P. Morgan.

Substrate: peer valuations

Valuations

Company	Ticker	Price (LCY\$)	PE (x)				PB (x)				NI yoy (%)			ROE (%)		
			2024	2025	2026E	2027E	2024	2025	2026E	2027E	2025	2026E	2027E	2025	2026E	2027E
Unimicron Tech	3037 TT	625	188.6	143.3	41.6	23.4	10.3	9.5	8.0	6.5	31%	244%	78%	7%	21%	30%
Kinsus	3189 TT	387	3,639.1	116.0	43.8	25.7	5.8	5.4	4.6	4.2	3020%	193%	74%	4%	11%	16%
Nan Ya Printed	8046 TT	669	2,090.6	255.3	65.9	39.2	9.5	9.5	8.6	8.0	753%	305%	78%	4%	14%	19%
Ibiden Co Ltd	4062 JP	9,810	81.3	36.0	48.8	42.8	5.6	4.7	4.4	4.0	126%	-26%	21%	14%	9%	10%
Samsung Electro	009150 KS	516,000	57.4	54.9	30.7	22.1	4.3	4.3	3.9	3.4	3%	79%	39%	8%	12%	15%

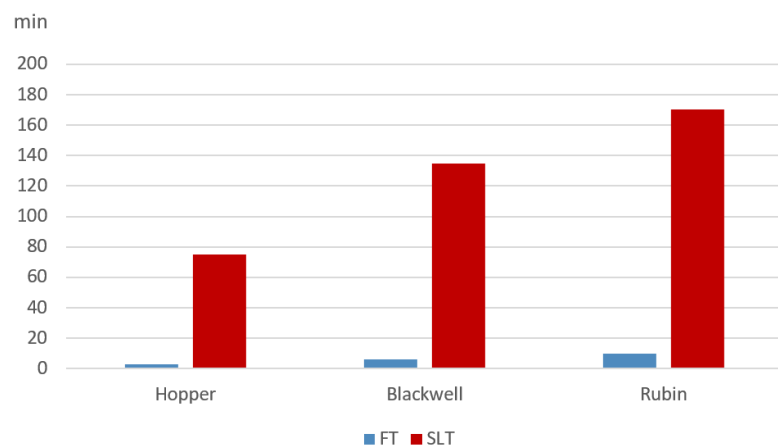
Note: 1) Share prices as of April 9; 2) J.P. Morgan covers Unimicron (covered by Jerry Tsai), Ibiden (by Akinori Kanemoto), and Samsung Electro (by Jay Kwon), whose financial forecasts are based on J.P. Morgan estimates. Others are based on Bloomberg Finance L.P. estimates.

Source: J.P. Morgan estimates, company data, Bloomberg Finance L.P.

Testing equipment

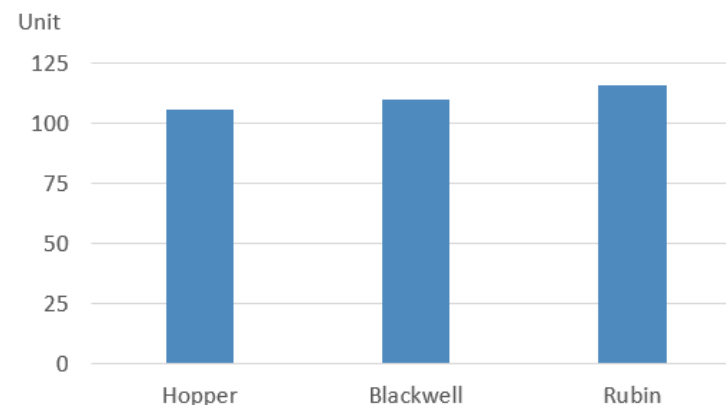
IC testing: 1) longer testing time (unit growth); 2) higher TDP (ASP growth)

Nvidia GPU FT/SLT testing time



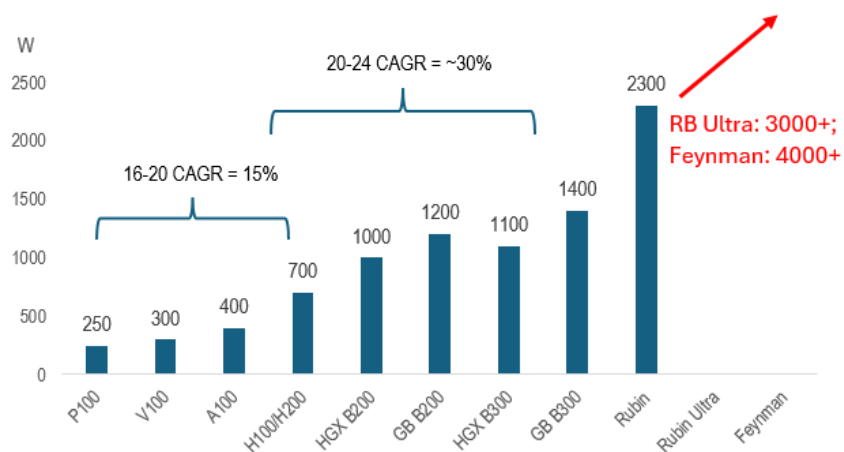
Source: J.P. Morgan estimates.

Nvidia GPU SLT equipment shipment (Chroma 100% share)



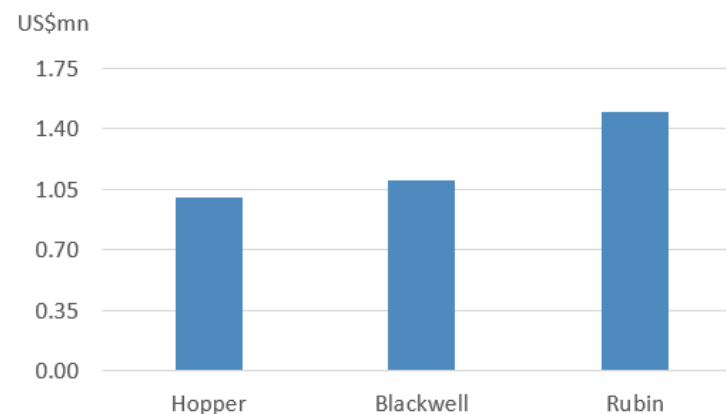
Source: J.P. Morgan estimates.

Nvidia GPU TDP per chip



Source: J.P. Morgan estimates.

Nvidia GPU SLT equipment ASP (Chroma 100% share)



Source: J.P. Morgan estimates.

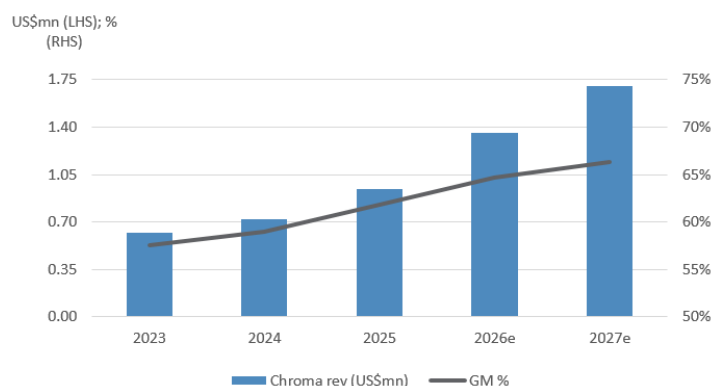
Chroma: SLT, Metrology and Datacenter Power Testing — each at 40%+ CAGR

Chroma segment overview

Rev weighting (%)		2023	2024	2025	2026E	2027E Notes
Testing rev (parent + overseas)		95%	95%	95%	97%	97%
Testing rev	Parent rev					
	ATS	43%	32%	33%	32%	33% Datacenter/AI power testing EV power testing Passive components, power adopters
	Semi/photon rev	16%	32%	33%	39%	41% SLT for GPU Metrology for CoWoS For laser emitters in fiber optical networks FT is final testers, for PMIC, MCU 3D sensing on iPhones; increase in 2022 and beyond from auto Lidar
	Turnkey	2%	3%	3%	7%	7% Turnkey is mostly for battery cell ; used to be Chinese battery makers, now more for non-Chinese
	Service & Others	6%	5%	5%	3%	3% Peripherals
Overseas		28%	22%	21%	15%	13% Include China battery formation business
MAS		4%	4%	4%	3%	2% Automation
New Material		0%	0%	0%	0%	0% Material trading (e.g., Copper, gold)
Others		1%	1%	1%	1%	1%

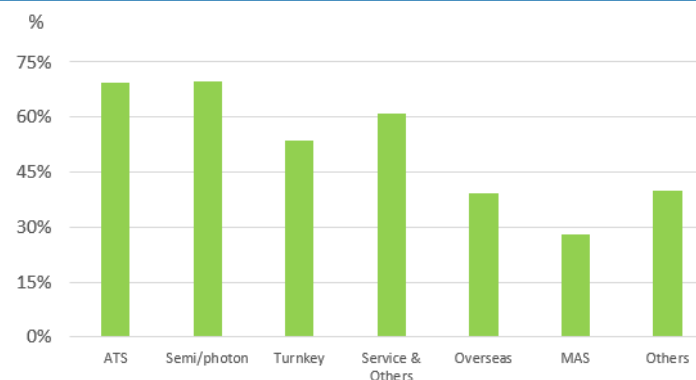
Source: J.P. Morgan estimates, company data.

Chroma rev. and GM



Source: J.P. Morgan estimates, company data.

2025 margin profile by segment

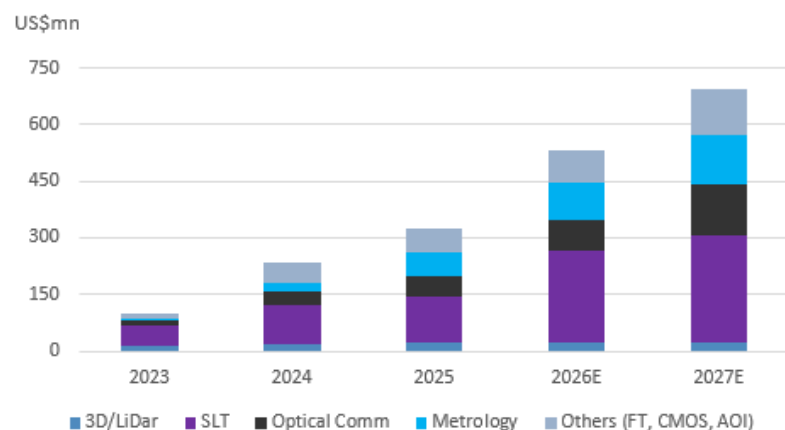


Source: J.P. Morgan estimates, company data.

J.P. Morgan

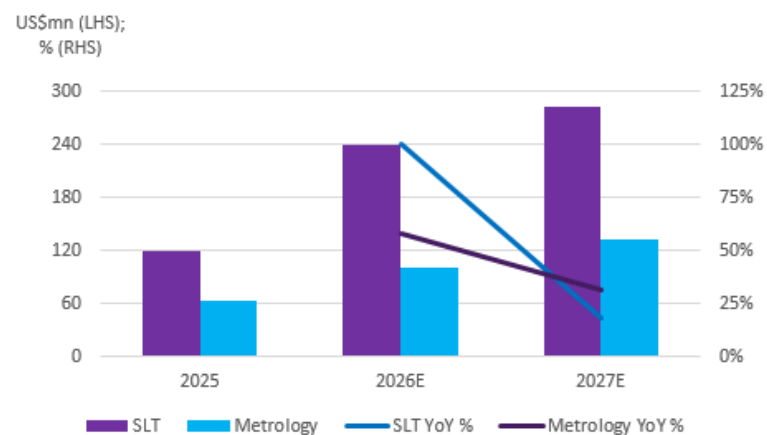
Chroma: SLT and metrology (both at 40%+ CAGR)

Semi/photon rev. mix



Source: J.P. Morgan estimates, company data.

SLT and metrology rev. growth



Source: J.P. Morgan estimates, company data.

Chroma SLT equipment vs peer

Chromal model 3260

Hon Prec HT-3015CT



Capacity (# of chip per batch)	6	6
Power (kW)	1.8	1.6
Temperature control	-40°C to 150°C	-55~130°C
ASP	US\$1-1.5mn	US\$8-1.1mn

Source: Company data.

Chroma ATC vs peer

Chromal 31000R/31000K

Hon Prec ATC 3.5

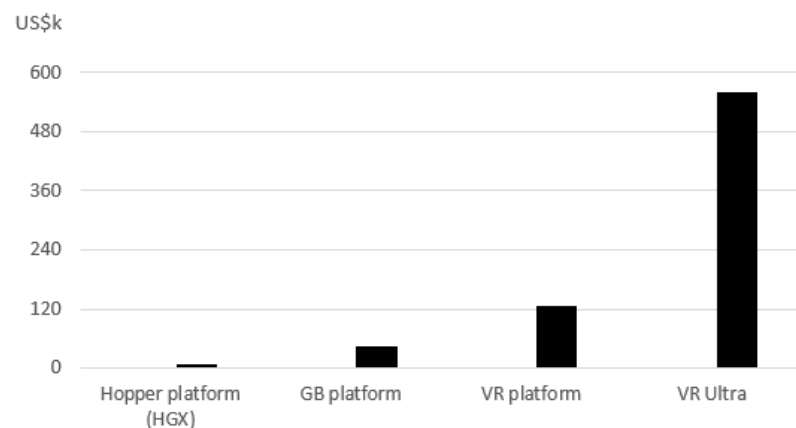


Power (kW)	2.9	2.0
Temperature control	-70°C to 150°C	25~125°C

Source: Company data.

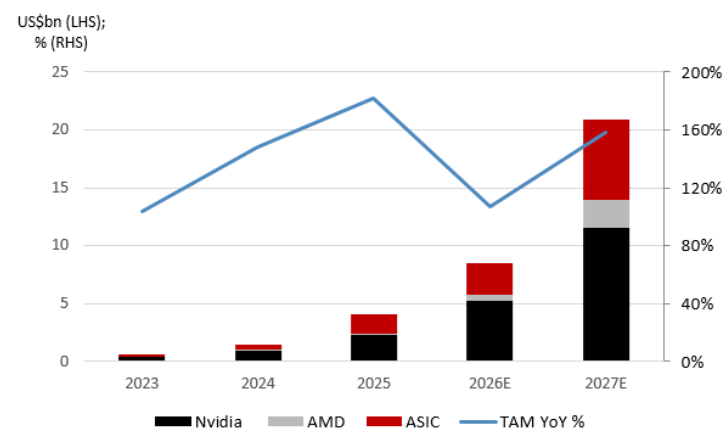
Chroma: power testing segment driven by data center power testing (TAM 80%+ CAGR)

Nvidia server power content per rack (PSU, etc)



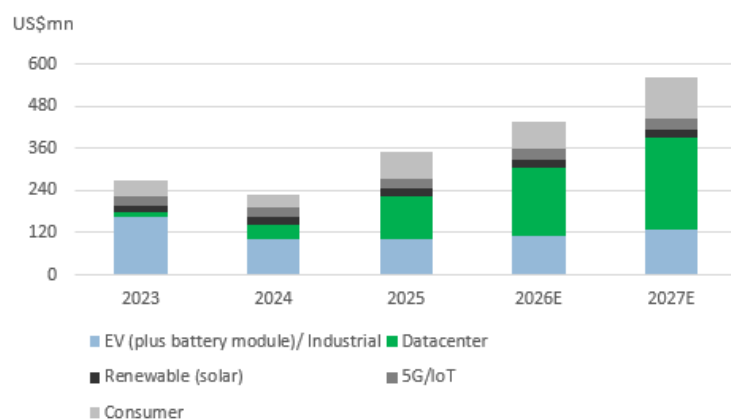
Source: J.P. Morgan estimates.

AI server power TAM (PSU, etc.)



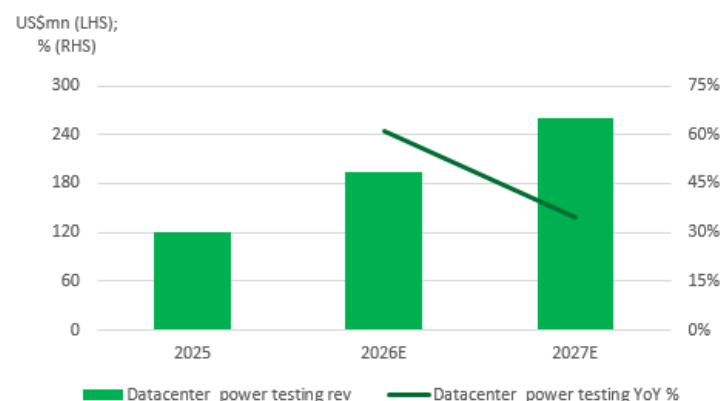
Source: Company data, J.P. Morgan estimates.

Chroma ATS/power testing rev. mix



Source: J.P. Morgan estimates, company data.

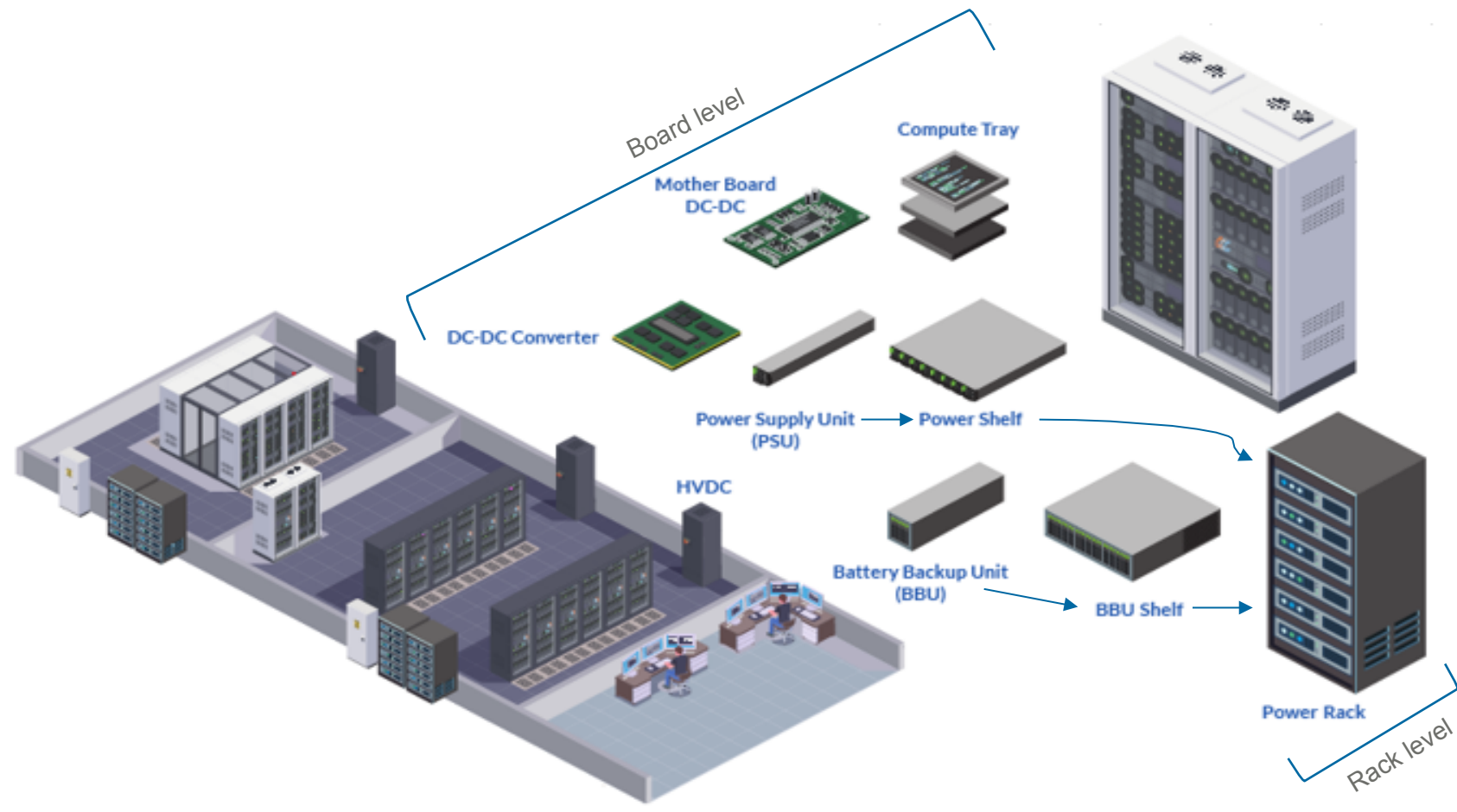
Chroma data center power testing rev. growth



Source: J.P. Morgan estimates, company data.

Overview of Data center Power Structure

- Chroma is the leader in AI data center power testing.
- Chroma has a comprehensive product offering in the datacenter power equipment testing realm (offering in each product category see next page).

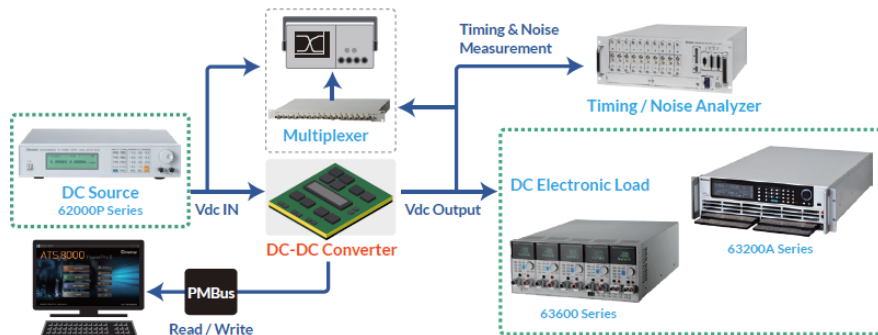


Source: Chroma.

J.P.Morgan

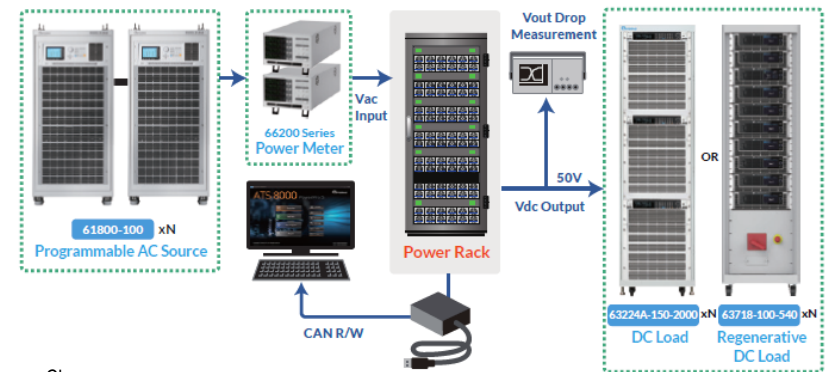
Chroma: datacenter power testing (under ATS segment) – comprehensive offering

DC converter testing product offering



Source: Chroma.

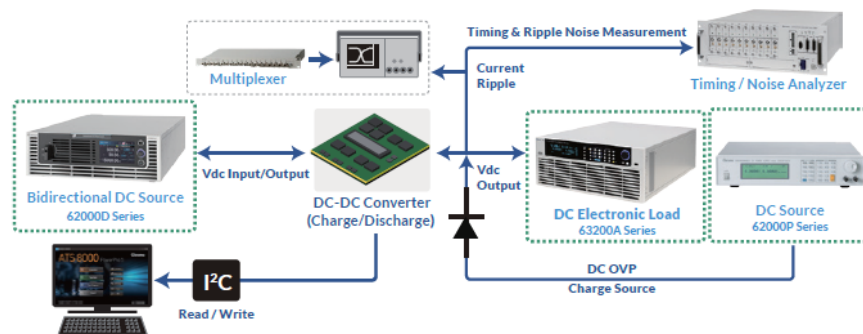
Rack level power testing product offering



Source: Chroma.

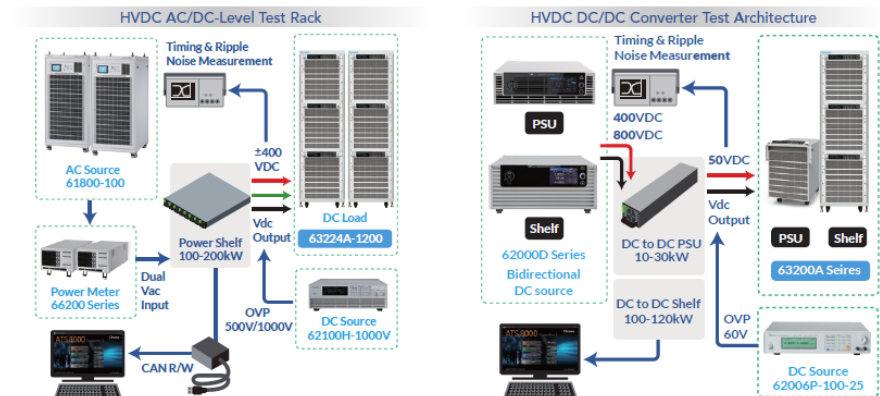
Future datacenter power supply technology

Back-up Battery Unit (BBU) testing product offering



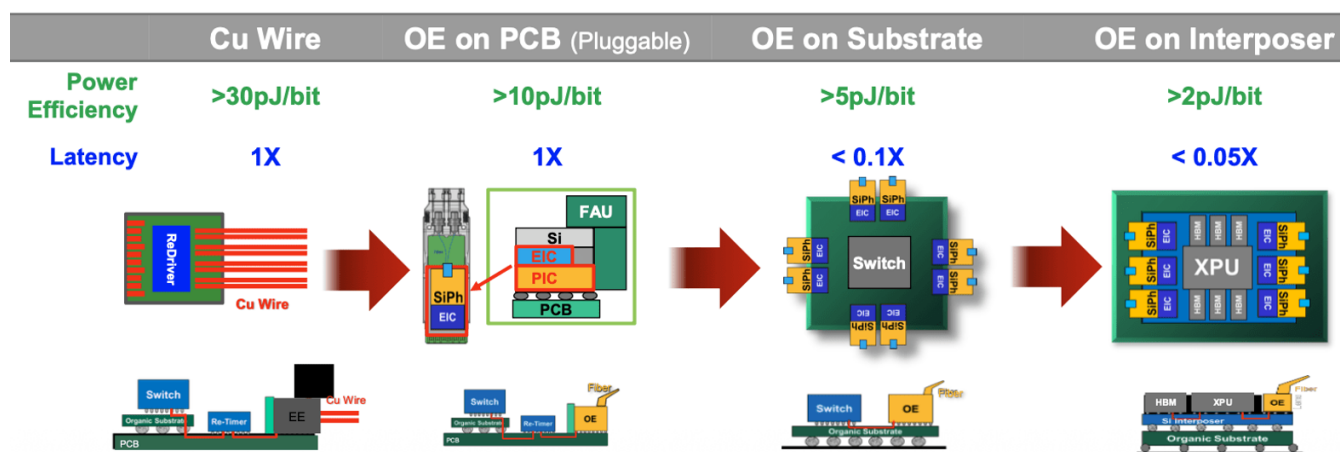
Source: Chroma.

HVDC power testing product offering



Source: Chroma.

Incremental catalyst: CPO testing



CPO testing process overview

Insertion	1	2	3	4
Process context	PIC Wafer-level  Photonic IC (modulator, power detector)	EIC-PIC Wafer-level  Electric IC on PIC wafer (driver, TIA)	Optical Engine Singulated Die/chip-level  Optical Engine (fiber attach, DSP, interposer)	ASIC/CPO Packaged module  ASIC with CPO (packaged module)
Notes	- single sided probing - electrical/optical direct current probing (power, loss, dark current etc)	- double sided probing - E/O, O/E, O/O	- full calibration (after RDL, DSP etc attached)	- full system functional validation (after co-packaged with ASIC)

Chroma

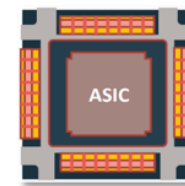


Chroma model 587 XX: insertion 3

Scale out tool calculation (Chroma 587 XX)

Number of racks	83,333
Each rack contains	6 ToR switch ICs (Spectrum)
Each switch IC contains	32 OEs
Total OE per year	16,000,000
Average daily testing volume	53,333 Assume 300 days
Average daily testing capacity required	96,000.0 Factor in 1.8x average

Spectrum: 32 OEs



ASIC + Optical Engine

Chroma tool capacity calculation

Hourly throughput	150
Daily output	3000 Assume 20 hours a day
Daily known-good-die	2400 Assume 80% yield

Tools needed **40**

Quantum 5 (NVLink switch): 3 OEs



ASIC + Optical Engine

GPU with CPO: 6 OEs

Testing equipment: peer valuations

Valuations

Company	Ticker	Price (LCY\$)	PE (x)				PB (x)				NI yoy (%)			ROE (%)		
			2024	2025	2026E	2027E	2024	2025	2026E	2027E	2025	2026E	2027E	2025	2026E	2027E
Chroma Ate	2360 TT	1,840	147.3	66.4	56.0	40.5	31.3	24.4	22.2	18.4	122%	19%	38%	41%	42%	50%
Hon Precision In	7769 TT	4,245	128.8	56.1	39.3	26.4	50.0	13.2	20.4	12.2	134%	54%	45%	34%	68%	71%
Advantest Corp	6857 JP	24,800	113.4	51.1	33.6	28.4	35.9	25.6	18.1	13.5	119%	52%	18%	58%	63%	10%
Teradyne Inc	TER US	364	106.8	104.7	61.5	45.9	20.9	20.3	N.A.	N.A.	2%	66%	32%	20%	31%	36%
Keysight Tec	KEYS US	323	91.5	65.5	35.5	30.1	10.9	9.4	7.9	6.5	38%	85%	17%	15%	24%	24%
Delta Elec	2308 TT	1,645	121.3	71.1	38.7	28.7	18.6	15.9	12.3	9.7	71%	84%	34%	24%	36%	38%

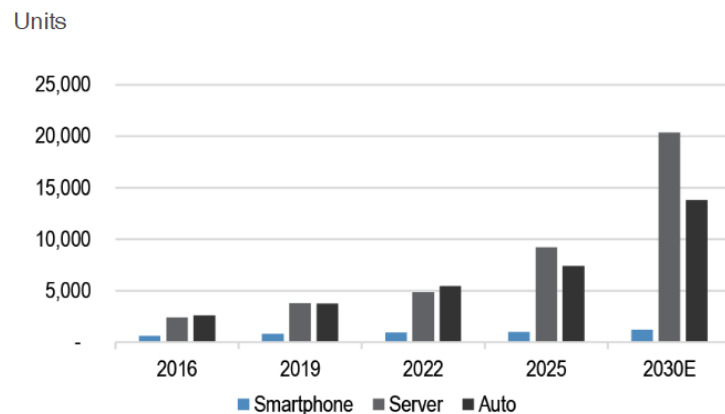
Note: 1) Share prices as of April 9; 2) J.P. Morgan covers Chroma ATE (covered by Jerry Tsai), Advantest (by Mio Shikanai), Teradyne (by Samik Chatterjee), Keysight (by Samik Chatterjee), and Delta Elec (by Albert Hung), whose financial forecasts are based on J.P. Morgan estimates. Others are based on Bloomberg Finance L.P. consensus.

Source: J.P. Morgan estimates, company data, Bloomberg Finance L.P.

Passive components

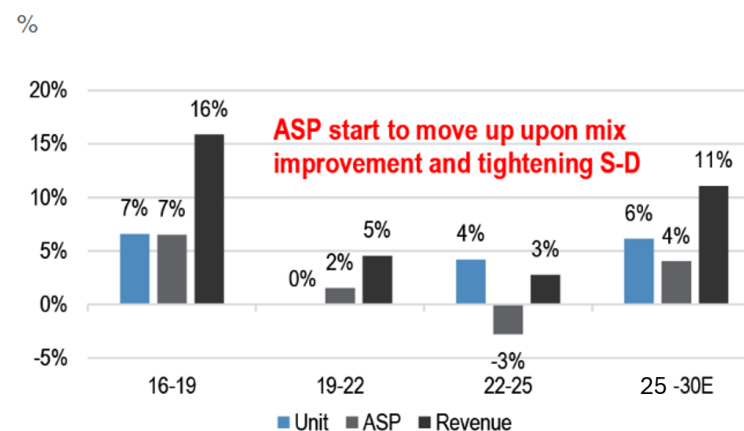
Passive component: 1) consumption increases from AI demand; 2) pricing stable

MLCC content comparison by application



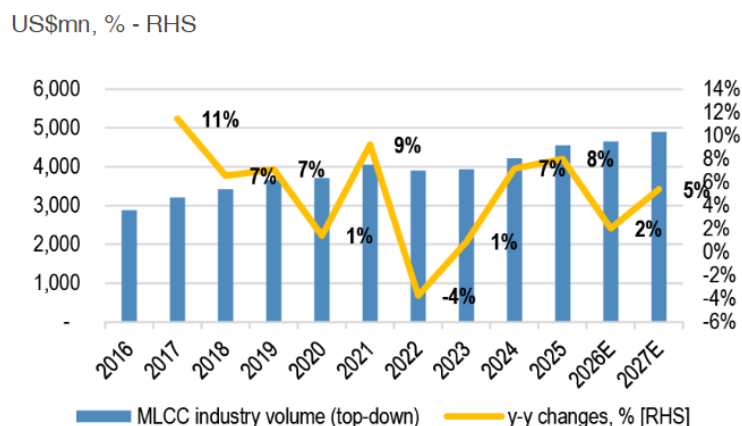
Source: J.P. Morgan estimates, company data.

MLCC industry revenue, volume, ASP CAGR comparison



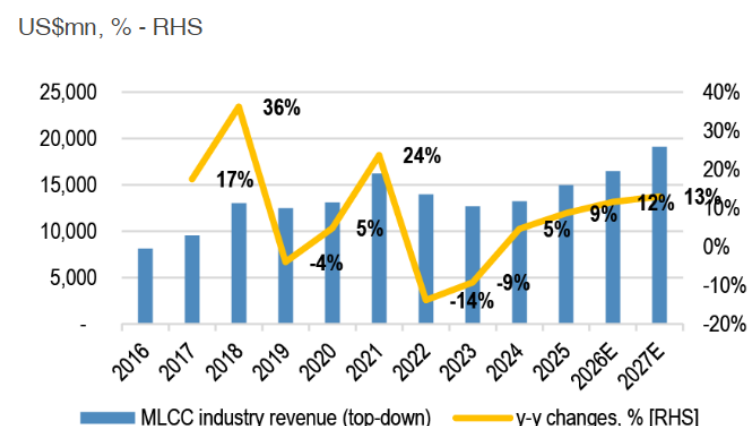
Source: J.P. Morgan estimates, company data.

MLCC industry volume and YoY



Source: J.P. Morgan estimates, company data.

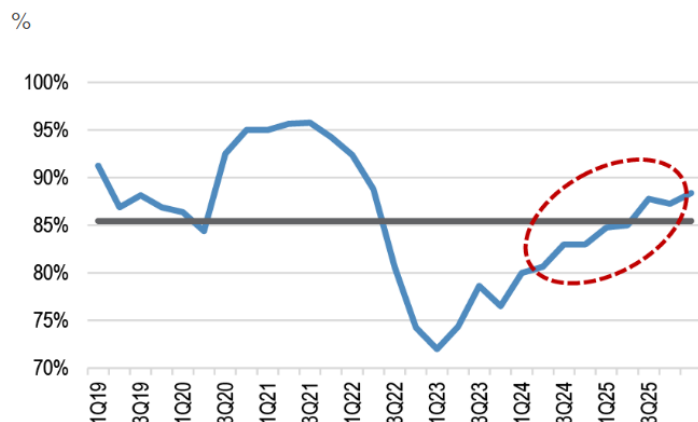
MLCC industry revenue and YoY



Source: J.P. Morgan estimates, company data.

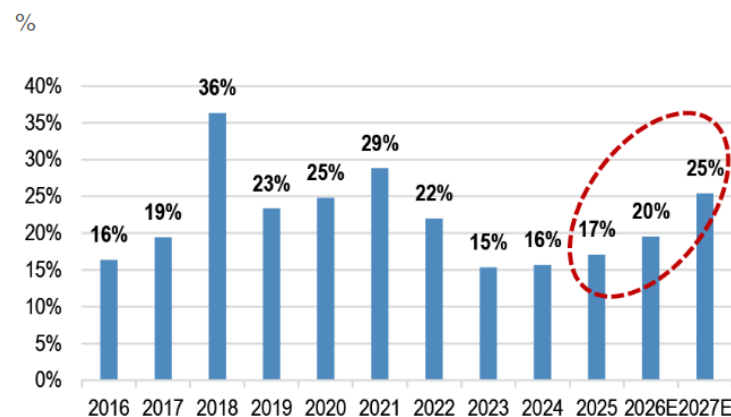
Passive component: 1) consumption increases from AI demand; 2) pricing stable

MLCC players' UTR trend



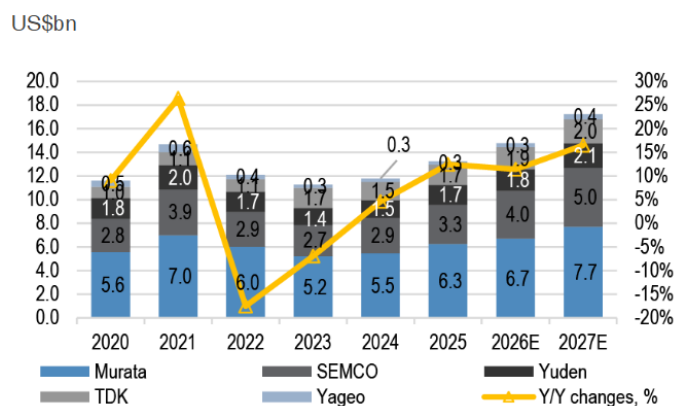
Source: J.P. Morgan estimates, company data.

MLCC industry bottom-up OPM trend



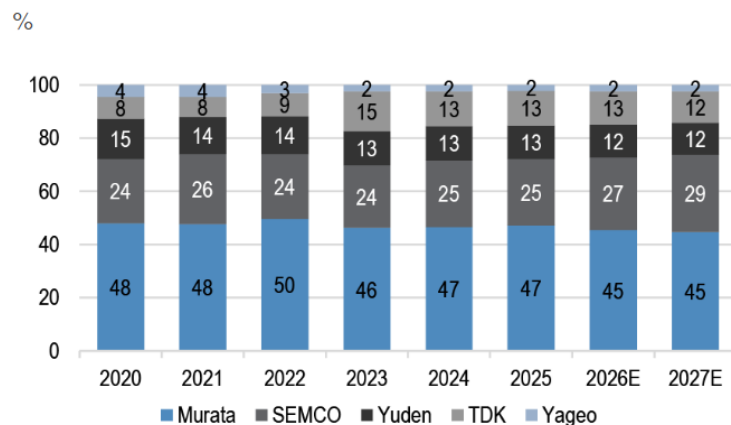
Source: J.P. Morgan estimates, company data.

MLCC industry bottom-up revenue and YoY trend



Source: J.P. Morgan estimates, company data.

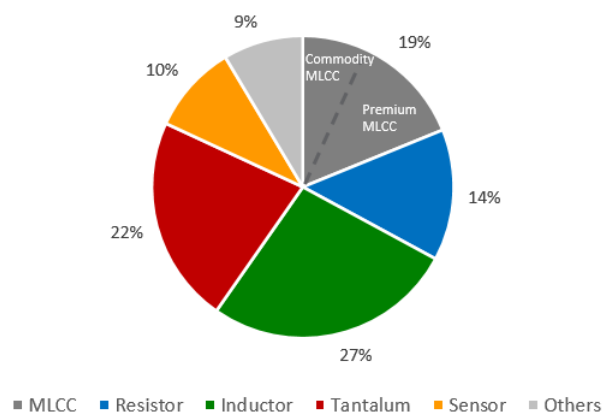
MLCC industry bottom-up share comparison (value)



Source: J.P. Morgan estimates, company data.

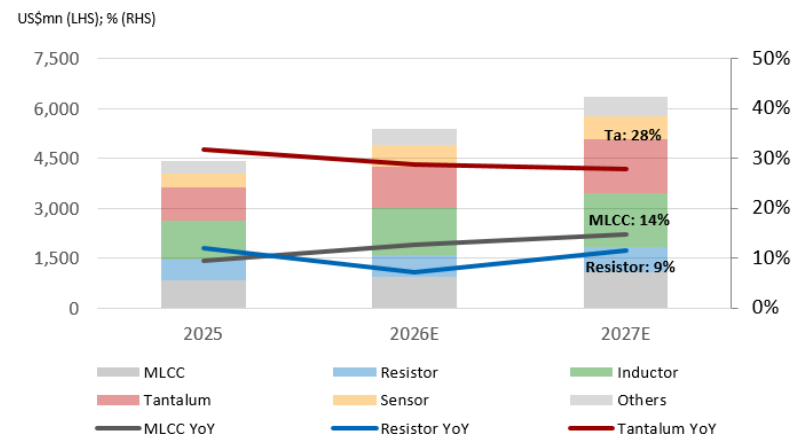
YAGEO: much more than MLCC ...Tantalum and resistor fuelling 20% rev CAGR (25-27E)

More than MLCC: 2025 rev. breakdown



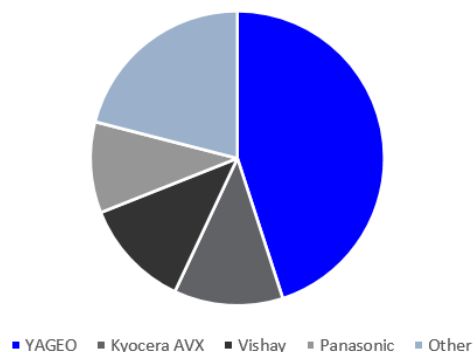
Source: J.P. Morgan estimates, company data.

YAGEO rev. growth by segment



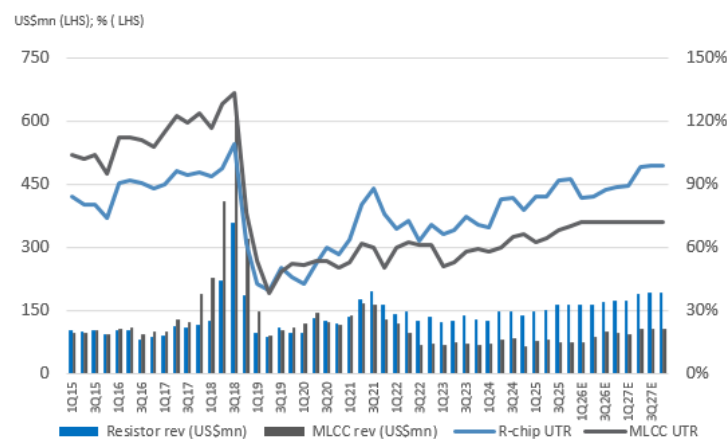
Source: J.P. Morgan estimates, company data.

Global Tantalum market share split



Source: J.P. Morgan estimates, company data.

Resistor rev. and UTR surpass MLCC (YAGEO m/s: 30-40%)



Source: J.P. Morgan estimates, company data.

Passive components: peer valuation

Valuations

Company	Ticker	Price (LCY\$)	PE (x)				PB (x)				NI yoy (%)			ROE (%)		
			2024	2025	2026E	2027E	2024	2025	2026E	2027E	2025	2026E	2027E	2025	2026E	2027E
Yageo	2327 TT	282	29.5	24.5	17.5	13.8	3.6	3.4	3.1	2.7	22%	41%	27%	14%	19%	21%
Murata Mfg Co	6981 JP	3,990	31.9	34.0	24.5	21.2	2.9	2.8	2.6	2.4	-9%	39%	47%	0%	11%	12%
Taiyo Yuden Co	6976 JP	4,863	260.5	37.9	25.3	22.1	1.9	1.9	1.8	1.8	588%	50%	25%	0%	7%	9%
Samsung Electro	009150 KS	516,000	57.4	55.2	30.7	22.1	4.3	3.9	3.9	3.4	4%	78%	39%	8%	12%	15%
Walsin Tech	2492 TT	124	20.1	26.1	16.5	N.A.	1.2	1.2	N.A.	N.A.	-23%	N.A.	N.A.	5%	N.A.	N.A.
Fenghua Advanced Technology	000636 CH	20	68.0	59.8	48.1	36.5	1.9	1.8	1.8	1.7	14%	24%	32%	3%	4%	5%
Vishay Intertech	VSH US	21	N.A.	N.A.	41.6	17.8	1.5	1.4	1.6	1.6	N.A.	N.A.	142%	0%	5%	N.A.

Note: 1) Share prices as of April 9; 2) J.P. Morgan covers YAGEO (covered by Jerry Tsai), Murata (by Akinori Kanemoto), Taiyo Yuden (by Akinori Kanemoto) and Samsung Electro (covered by Jay Kwon), whose financial forecasts are based on J.P. Morgan estimates. Others are based on Bloomberg Finance L.P. consensus.

Source: J.P. Morgan estimates, company data, Bloomberg Finance L.P.

Disclosures

Analyst Certification: The Research Analyst(s) denoted by an “AC” on the cover of this report certifies (or, where multiple Research Analysts are primarily responsible for this report, the Research Analyst denoted by an “AC” on the cover or within the document individually certifies, with respect to each security or issuer that the Research Analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect the Research Analyst’s personal views about any and all of the subject securities or issuers; and (2) no part of any of the Research Analyst’s compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the Research Analyst(s) in this report. For all Korea-based Research Analysts listed on the front cover, if applicable, they also certify, as per KOFIA requirements, that the Research Analyst’s analysis was made in good faith and that the views reflect the Research Analyst’s own opinion, without undue influence or intervention.

All authors named within this report are Research Analysts who produce independent research unless otherwise specified. In Europe, Sector Specialists (Sales and Trading) may be shown on this report as contacts but are not authors of the report or part of the Research Department.

Important Disclosures

Company-Specific Disclosures: J.P. Morgan does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. Important disclosures, including price charts and credit opinion history tables (if applicable), are available for compendium reports and all J.P. Morgan–covered companies, and certain non-covered companies, by visiting <https://www.jpmm.com/research/disclosures>, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

Explanation of Equity Research Ratings, Designations and Analyst(s) Coverage Universe:

J.P. Morgan uses the following rating system: Overweight (over the duration of the price target indicated in this report, we expect this stock will outperform the average total return of the stocks in the Research Analyst’s, or the Research Analyst’s team’s, coverage universe); Neutral (over the duration of the price target indicated in this report, we expect this stock will perform in line with the average total return of the stocks in the Research Analyst’s, or the Research Analyst’s team’s, coverage universe); and Underweight (over the duration of the price target indicated in this report, we expect this stock will underperform the average total return of the stocks in the Research Analyst’s, or the Research Analyst’s team’s, coverage universe). NR is Not Rated. In this case, J.P. Morgan has removed the rating and, if applicable, the price target, for this stock because of either a lack of a sufficient fundamental basis or for legal, regulatory or policy reasons. The previous rating and, if applicable, the price target, no longer should be relied upon. An NR designation is not a recommendation or a rating. Some stocks under coverage have a rating but no price target; in these cases, we expect the stock will outperform/perform in line/underperform the average total return of the stocks in the Research Analyst’s, or the Research Analyst’s team’s, coverage universe of the relevant duration of the region. In our Asia (ex-Australia and ex-India) and U.K. small- and mid-cap Equity Research, each stock’s expected total return is compared to the expected total return of a benchmark country market index, not to those Research Analysts’ coverage universe. If it does not appear in the Important Disclosures section of this report, the certifying Research Analyst’s coverage universe can be found on J.P. Morgan’s Research website, <https://www.jpmorganmarkets.com>.

Coverage Universe: Tsai, Jerry : Chicony Power Technology (6412.TW), Chroma ATE (2360.TW), Chunghwa Precision Test Tech (6510.TWO), E Ink (8069.TWO), Elite Material Co (2383.TW), Flexium Interconnect Inc (6269.TW), Radiant Opto-Electronics Corp. (6176.TW), Sunonwealth (2421.TW), Taiwan Surface Mount Tech (6278.TW), Tripod Technology Corp (3044.TW), Unimicron (3037.TW), WIN Semiconductors Corp (3105.TWO), YAGEO (2327.TW)

Disclosures

J.P. Morgan Equity Research Ratings Distribution, as of April 04, 2026

	Overweight (buy)	Neutral (hold)	Underweight (sell)
J.P. Morgan Global Equity Research Coverage*	51%	37%	12%
IB clients**	83%	79%	74%
JPMS Equity Research Coverage*	49%	39%	13%
IB clients**	94%	93%	85%

*Please note that the percentages may not add to 100% because of rounding.

**Percentage of subject companies within each of the "buy," "hold" and "sell" categories for which J.P. Morgan has provided investment banking services within the previous 12 months.

For purposes of FINRA ratings distribution rules only, our Overweight rating falls into a buy rating category; our Neutral rating falls into a hold rating category; and our Underweight rating falls into a sell rating category. Please note that stocks with an NR designation are not included in the table above. This information is current as of the end of the most recent calendar quarter.

Equity Valuation and Risks: For valuation methodology and risks associated with covered companies or price targets for covered companies, please see the most recent company-specific research report at <http://www.jpmorganmarkets.com>, contact the primary analyst or your J.P. Morgan representative, or email research.disclosure.inquiries@jpmorgan.com. For material information about the proprietary models used, please see the Summary of Financials in company-specific research reports and the Company Tearsheets, which are available to download on the company pages of our client website, <http://www.jpmorganmarkets.com>. This report also sets out within it the material underlying assumptions used.

History of Investment Recommendations:

A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of <http://www.jpmorganmarkets.com> where you can also search by analyst name, sector or financial instrument.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Registration of non-US Analysts: Unless otherwise noted, the non-US analysts listed on the front of this report are employees of non-US affiliates of J.P. Morgan Securities LLC, may not be registered as research analysts under FINRA rules, may not be associated persons of J.P. Morgan Securities LLC, and may not be subject to FINRA Rule 2241 or 2242 restrictions on communications with covered companies, public appearances, and trading securities held by a research analyst account.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to [UK MIFID Research Unbundling exemption](#) for details of J.P. Morgan's implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

All research material made available to clients are simultaneously available on our client website, J.P. Morgan Markets, unless specifically permitted by relevant laws. Not all research content is redistributed, e-mailed or made available to third-party aggregators. For all research material available on a particular stock, please contact your sales representative.

J.P.Morgan

Disclosures

Any long form nomenclature for references to China; Hong Kong; Taiwan; and Macau within this research material are Mainland China; Hong Kong SAR (China); Taiwan (China); and Macau SAR (China).

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Any digital or crypto assets discussed in this research report are subject to a rapidly changing regulatory landscape. For relevant regulatory advisories on crypto assets, including bitcoin and ether, please see <https://www.jpmorgan.com/disclosures/cryptoasset-disclosure>.

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC (“JPMS”) acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit <https://www.theocc.com/components/docs/riskstoc.pdf> for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or https://www.finra.org/sites/default/files/2020-08/Security_Futures_Risk_Disclosure_Statement_2020.pdf for a copy of the Security Futures Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries (“J.P. Morgan Private Bank”), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Where more than one legal entity is listed under an analyst’s name, the first legal entity is responsible for the production unless stated otherwise. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below in the legal entity disclosures, this material has been distributed by the legal entity responsible for production, or where more than one legal entity is listed under the analyst’s name, the first legal entity will be responsible for distribution. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Disclosures

Legal Entities Disclosures and Country-/Region-Specific Disclosures:

Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina (“BCRA”- Central Bank of Argentina) and Comisión Nacional de Valores (“CNV”- Argentinian Securities Commission - ALYC y AN Integral N°51).

Australia: J.P. Morgan Securities Australia Limited (“JPMSAL”) (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to "wholesale clients" (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting <https://www.jpmm.com/research/disclosures>. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia’s Research Independence Policy which can be found at the following link: [J.P. Morgan Australia - Research Independence Policy](#).

Brazil: Banco J.P. Morgan S.A. is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmchase.com.

Canada: J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P.Morgan Securities Canada Inc.

Chile: Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile.

China: J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business.

Colombia: Banco J.P. Morgan Colombia S.A. is supervised by the Superintendencia Financiera de Colombia (SFC).

Dubai International Financial Centre (DIFC): JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules.

European Economic Area (EEA): Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE (“JPM SE”), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions (“EEA professional investors”). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons.

Hong Kong: J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited.

India: J.P. Morgan India Private Limited (Corporate Identity Number - U67120MH1992FTC068724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai – 400098, is registered with the Securities and Exchange Board of India (SEBI) as a ‘Research Analyst’ having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number – INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: <http://www.jpmypl.com>. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license

Disclosures

allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Prasanna Bandal; prasanna.bandal@jpmchase.com; +912261575159. Grievance Officer: Ramprasad K, jpmipl.research.feedback@jpmorgan.com; +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Please visit [Terms and Conditions and Most Important Terms and Conditions \(MITC\)](#). The annual Compliance audit report is available at <http://www.jpmipl.com/#research>.

Indonesia: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK).

Korea: J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch.

Japan: JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan.

Malaysia: This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia.

Mexico: J.P. Morgan Casa de Bolsa, S.A. de C.V. and J.P. Morgan Grupo Financiero are members of the Mexican Stock Exchange and are authorized to act as a broker dealer by the National Banking and Securities Exchange Commission.

New Zealand: This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008.

Philippines: J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission.

Singapore: This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMS) [MDDI (P) 057/08/2025 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of "accredited investors," "expert investors" or "institutional investors," as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material.

South Africa: J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA).

Taiwan: J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. **To the extent that J.P. Morgan Securities (Taiwan) Limited produces research materials on securities not listed on the Taiwan Stock Exchange or Taipei Exchange ("Non-Taiwan Listed Securities"), these materials shall not constitute securities recommendations for the purpose of applicable Taiwan regulations, and, for the avoidance of doubt, J.P. Morgan Securities (Taiwan) Limited does not act as broker for Non-Taiwan Listed Securities.** According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the "Important Disclosures" in this material.

Thailand: This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission, and its registered address is 3rd Floor, 20 North Sathorn Road, Silom, Bangrak, Bangkok 10500.

UK: Research is produced in the UK by J.P. Morgan Securities plc ("JPMS plc") which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority or J.P. Morgan Markets Limited ("JPMML Ltd") which is authorised and regulated by the Financial Conduct Authority. Unless specified to the contrary, this material is distributed in the UK by JPMS plc and is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 ("the FPO"); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made;

Disclosures

all such persons being referred to as "UK relevant persons". This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. A description of J.P. Morgan EMEA's policy for prevention and avoidance of conflicts of interest related to the production of Research can be found at the following link: [J.P. Morgan EMEA - Research Independence Policy](#).

U.S.: J.P. Morgan Securities LLC ("JPMS") is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. Artificial intelligence tools may have been used in the preparation of this material, including assisting in data analysis, pattern recognition, and content drafting for research material. J.P. Morgan accepts no liability whatsoever for any loss arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and responsibilities that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results. Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: <https://www.jpmorgan.com/disclosures/email>

Disclosures

MSCI: Certain information herein (“Information”) is reproduced by permission of MSCI Inc., its affiliates and information providers (“MSCI”) ©2026. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to [msci.com/disclaimer](https://www.msci.com/disclaimer)

Sustainalytics: Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or its use. The use of the data is subject to conditions available at <https://www.sustainalytics.com/legal-disclaimers>. ©2026 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised April 04, 2026.

Copyright 2026 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. It is strictly prohibited to use or share without prior written consent from J.P. Morgan any research material received from J.P. Morgan or an authorized third-party (“J.P. Morgan Data”) in any third-party artificial intelligence (“AI”) systems or models when such J.P. Morgan Data is accessible by a third-party.